

Lecture Notes: Mechanical Workshop Practice – Conventional Machines

1. Introduction to Conventional Machines

Conventional machines are manually or mechanically operated tools used in workshops to shape, cut, drill, or finish raw materials (metal, wood, plastic). They form the foundation of manufacturing and require hands-on operator skill. Key machines include:

- Lathe
- Drilling Machine
- Milling Machine
- Grinding Machine
- Shaper Machine
- Planer Machine
- Power Hacksaw

2. Lathe Machine

Purpose: For cylindrical turning, facing, drilling, and threading.

Key Components:

- Bed: Rigid base for alignment.
- Headstock: Holds spindle and gears for rotation.
- Tailstock: Supports workpiece for drilling/turning.
- Carriage: Moves cutting tool longitudinally or crosswise.

Common Operations:

- Turning: Removing material to create cylindrical shapes.
- Facing: Creating flat surfaces on the workpiece end.
- Taper Turning: Producing conical shapes.
- Thread Cutting: Using a lead screw and gears.

Safety:

- Secure workpiece in chuck.
- Avoid loose clothing near rotating parts.

3. Drilling Machine

Purpose: Creating holes in materials.

Types:

- Bench Drill: For small workpieces.
- Radial Drill: Adjustable arm for large components.
- Pillar Drill: Vertical drilling with precision.

Key Components:

- Spindle: Rotates the drill bit.
- Feed Handle: Controls downward movement.
- Drill Chuck: Holds the drill bit.

Operations:

- Drilling: Basic hole creation.
- Counterboring: Enlarging hole openings.
- Reaming: Finishing holes to precise dimensions.

Safety:

- Clamp workpiece firmly.
- Wear safety goggles to prevent chip injuries.

4. Milling Machine

Purpose: Cutting slots, gears, and complex profiles.

Types:

- Horizontal Milling: Uses arbor-mounted cutters.
- Vertical Milling: Spindle moves vertically for face milling.

Key Components:

- Knee: Adjusts vertical height.
- Table: Holds workpiece and moves in X/Y axes.
- Arbor: Holds milling cutters.

Operations:

- Face Milling: Producing flat surfaces.
- Slotting: Cutting grooves.
- Gear Cutting: Using form cutters.

Safety:

- Use vise to secure workpiece.
- Keep hands clear of rotating cutters.

5. Grinding Machine

Purpose: Precision finishing and sharpening tools.

Types:

- Surface Grinder: For flat surfaces.
- Cylindrical Grinder: For round components.
- Tool & Cutter Grinder: Sharpens drills, milling cutters.

Key Components:

- Grinding Wheel: Made of abrasive grains (e.g., aluminum oxide).
- Coolant System: Prevents overheating.

Operations:

- Surface Grinding: Achieving smooth finishes.
- Tool Sharpening: Restoring cutting edges.

Safety:

- Inspect grinding wheel for cracks.
- Use eye protection and face shield.

6. Shaper Machine

Purpose: Producing flat surfaces, grooves, and keyways.

Key Components:

- Ram: Reciprocates the cutting tool.
- Clapper Box: Lifts tool during return stroke.
- Operations:
 - Horizontal Cutting: Flat surfaces.
 - Vertical Cutting: Slots or keyways.

Safety:

- Keep hands away from moving ram.

- Use guards to contain chips.

7. Planer Machine

Purpose: Machining large flat surfaces (similar to shaper but for bigger workpieces).

Key Difference: Workpiece moves; tool remains stationary.

Operations:

- Planning: Heavy-duty flat surface finishing.

8. Power Hacksaw

Purpose: Cutting metal bars, pipes, or rods.

Key Components:

- Reciprocating Blade: High-speed steel teeth.
- Vice: Holds workpiece.

Operations:

- Straight Cutting: Sizing raw materials.
- Angular Cutting: Adjusting vice for angled cuts.

Safety:

- Ensure blade tension is correct.
- Secure workpiece tightly.

Safety Guidelines for Conventional Machines

1. Personal Protective Equipment (PPE): Safety glasses, gloves, and aprons.
2. Workpiece Securing: Always clamp materials to avoid slippage.
3. Machine Guards: Use shields to protect against flying chips.
4. Power Off: Disconnect machines during tool changes or adjustments.

Summary Table

Machine	Primary Use**	Key Operations
Lathe	Cylindrical shaping	Turning, Facing, Threading
Drilling	Hole-making	Drilling, Reaming, Counterboring
Milling	Complex profiles	Slotting, Gear Cutting
Grinding	Surface finishing	Tool Sharpening, Precision Finishing
Shaper	Flat surfaces & keyways	Horizontal/Vertical Cutting
Power Hacksaw	Metal cutting	Straight/Angular Cuts

Key Takeaways

1. Conventional machines require manual skill and adherence to safety protocols.
2. Each machine has specific applications: lathe for cylindrical work, milling for complex shapes, grinding for finishing.
3. Regular maintenance (e.g., lubrication, blade sharpening) ensures machine longevity.

Practical Tips:

- Always use sharp tools for efficient cutting.
- Measure twice before machining to minimize errors.
- Start with low speeds for unfamiliar operations.

For hands-on training, practice under supervision and refer to workshop manuals for machine-specific instructions.